

PROFESSIONAL SUMMARY

Robotics Engineer specializing in **ROS2 autonomous navigation, SLAM, sensor fusion, and vision-guided UAV systems**. Proven track record of designing, simulating, and hardware-testing autonomous mobile robots (AMRs) and multirotor drones from mathematical modeling and Gazebo SITL to field deployment. **Top 10 / 400+ teams** at IIT Bombay e-Yantra 2025–26. Skilled in ROS2 Humble, ArduPilot, PX4, MAVSDK, Pixhawk controllers, OpenCV, YOLO, and embedded hardware development.

TECHNICAL SKILLS

Autonomous & Controls:	Autonomous Systems, PID Control, Sensor Fusion, EKF (Extended Kalman Filter), RANSAC, ROS2 Control (ros2_control), Visual Servoing, Trajectory Planning, Kinematics
UAV & Multirotors:	MultiRotor UAVs, Pixhawk Flight Controller, ArduPilot, PX4 Autopilot, QGroundControl, Mission Planner, MAVLink, MAVSDK, DroneKit
Robotics & SLAM:	ROS2 (Humble/Foxy), Nav2 Stack, MoveIt2, SLAM Toolbox, ORB-SLAM3, AMCL, TF2 Transformation Tree, URDF / xacro
Perception & Vision:	OpenCV, YOLOv8 / YOLOv4-Tiny, Raspberry Pi Cam v3 NoIR, ESP-CAM, ArUco Marker Geolocation, Pinhole Camera Model
Languages & Concurrency:	C++, Python, Multithreading, Async I/O, Gazebo, MuJoCo, RViz2, ArduPilot SITL, MATLAB / Simulink
Embedded & Compute Hardware:	NVIDIA Jetson Nano & AGX Xavier, Raspberry Pi 4/5, ESP32-S3 / ESP-CAM, Microprocessors, Arduino, RPLidar A1M8, LiDAR, MPU9250 / MPU6050 IMU, Encoder Motors, Barometer, NRF24L01, LoRa Module, ST7920 Display, Custom MOSFET ESC
CAD, Tools & OS:	Fusion 360, 3D Printing, TinkerCad, Git, Linux (Ubuntu 22.04)

EDUCATION

PDPM Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur

Bachelor of Technology (B.Tech) — Electronics and Communication Engineering

Coursework:

Robotics (Kinematics, D-H Parameters, Trajectory Planning), Control Systems, Embedded Systems, Sensors & Actuators.

Practical Labs:

PUMA Robot Kinematics, Trajectory-Following Autonomous Mobile Robots.

Aug 2024 – Aug 2028

LEADERSHIP & EXPERIENCE

Coordinator — Electronics and Robotics Society (ERS)

PDPM IIITDM Jabalpur

Led technical program for 200+ members; organized 10+ hands-on robotics workshops per semester on ROS2 and embedded systems.

Mentored 30+ junior students in ROS2, SLAM, and autonomous systems; coordinated team participation for e-Yantra at IIT Bombay.

Aug 2024 – Present

Core Committee Member — Aero Fabrication Club

PDPM IIITDM Jabalpur

Designed and fabricated UAV hardware (frame assembly, ESC wiring, Pixhawk flight controller integration) for national competitions.

Aug 2024 – Present

FEATURED ROBOTICS PROJECTS

e-Yantra KrishiCobot — IIT Bombay (e-YRC 2025–26)

[GitHub]

ROS2 Humble · LiDAR · IMU · OpenCV · ArUco · MoveIt2 · UR5 Arm · Visual Servoing

Engineered a fully autonomous agricultural robot for crop monitoring and fruit harvesting; achieved **Top 10 / 400+ teams**.

Built a multi-sensor perception pipeline (LiDAR + ultrasonic + IMU) enabling collision-free lane navigation across field rows.

Developed ArUco marker-based fruit classification and automated pick-and-place routines with a UR5 arm via MoveIt2 and visual servoing.

Aug 2025 – May 2026

WareOps — Autonomous Warehouse Auditing Robot & Active Vision System

[GitHub Org]

ROS2 Humble · Nav2 · AMCL · SLAM Toolbox · ros2_control · RPLidar A1M8 · OpenCV · PyZbar · Gazebo Ignition · ESP32-CAM

Engineered an end-to-end AMR system for automated warehouse auditing: autonomous rack-to-rack transit using Nav2, AMCL localization, and SLAM Toolbox occupancy mapping.

Built a 2-DOF active vision scanning mast with vertical tilt & horizontal pan gimbal sweeps; dual QR pipeline (OpenCV WeChat QR + PyZbar) for reliable multi-tier rack scanning.

Achieved full sim-hardware parity: Gazebo Ignition warehouse environment (custom SDF + URDF Xacro) mirroring the hardware stack — RPi 4, RPLidar A1M8, Arduino Nano via ros2_control, and ESP32-CAM WiFi bridge.

Dec 2025 – Aug 2026

Disaster Surveillance Drone — NIDAR 2026

[GitHub]

Python · MAVSDK · ArduPilot SITL · MAVLink · QGroundControl · Mission Planner · YOLOv8

Architected a polygon-based autonomous survey UAV covering **75+ acres in <20 minutes** at 25m altitude via optimized zigzag mission paths.

Integrated YOLOv8 real-time human detection with Return-To-Launch (RTL) failsafe, geofencing, and pause-center-resume target lock-on.

Jun 2025 – Jan 2026

- Fully automated mission lifecycle (upload → arm → takeoff → survey → RTL) via MAVSDK Python API, reducing operator workload to zero.

[Autonomous Drone Survey & Target Detection — SAE Aerothon 2025](#) [\[GitHub\]](#)

Python · ROS2 · DroneKit · Pixhawk · ArduPilot · MAVLink · Gazebo · YOLOv4-Tiny · PID Control

- Assembled a ROS2 autonomous drone for survey and target detection; selected for national-level competition presentation.
- Implemented pinhole camera model pixel-to-GPS conversion enabling sub-meter target geolocation from aerial imagery.
- Engineered a visual PID centering loop for precise autonomous target alignment and lock-on during hover operations.

[Differential Drive Robot — ROS2 Navigation & SLAM](#) [\[GitHub\]](#) (★ 7 Stars)

C++ · Python · ROS2 Humble · Gazebo · Nav2 · SLAM Toolbox · AMCL · URDF/xacro · EKF

- Modeled and simulated a 4-wheeled differential drive robot with URDF/xacro physics model in Gazebo environment.
- Achieved SLAM-based 2D occupancy mapping using SLAM Toolbox and AMCL pose tracking for GPS-denied indoor localization.
- Deployed fully autonomous path planning, obstacle avoidance, and goal navigation using Nav2 stack with EKF sensor fusion.

[ESP32-S3 Quadcopter Drone](#) [\[YouTube\]](#)

ESP32-S3 · MPU6050 IMU · Barometer · Custom MOSFET ESC · Embedded C · TinkerCad

- Fabricated a quadcopter drone from scratch with IMU stabilization and barometer altitude sensing.
- Designed a custom MOSFET-based Electronic Speed Controller (ESC) and flashed custom flight control firmware in Embedded C.

[Modular Robotic Arm Platform](#) [\[GitHub\]](#)

Servo Motors · Arduino · Kinematics · Trajectory Planning · 3D CAD

- Developed a 5-DOF modular robotic arm with inverse kinematics for automated pick-and-place operations.

KEY ACHIEVEMENTS & CERTIFICATIONS

 **e-Yantra Robotics (IIT Bombay):** Top 10 / 400+ Teams ([Certificate](#))

 **DD Robocon 2025:** Stage-2 Qualification ([Certificate](#))

 **SAE Aerothon 2025:** Phase 2 Certificate ([Certificate](#))

 **NIDAR 2026:** National UAV Challenge ([Certificate](#))